Exercises

Problem 6.1: Ramsey measurement

Consider a single qubit with energy splitting ω and Hamiltonian $\hat{H}$

$$\hat{H} = \frac{1}{2}\omega\hat{\sigma}_z. \tag{6.1}$$

We consider a Ramsey measurement for estimating ω. The Ramsey sequence consists of the steps: *(i)* initialise the qubit in state $|0\rangle$, *(ii)* apply the rotation $\hat{R}_y(\frac{\pi}{2})$, i.e. a $\frac{\pi}{2}$ pulse (c.f. Eq. (2.53)), *(iii)* let the qubit evolve freely under $\hat{H}$ for a time t, *(iv)* apply the inverse rotation $\hat{R}_y(-\frac{\pi}{2})$, and *(v)* measure in the $\{|0\rangle, |1\rangle\}$ basis.

(a) Illustrate each step *(i)-(v)* on the Bloch sphere.

(b) Show that the probability of the outcome corresponding to $|1\rangle$ is given by

$$p_1 = \sin^2(\frac{\omega t}{2}). \tag{6.2}$$

(c) Show that the Fisher information of the distribution of outcomes, with respect to ω, is

$$\mathcal{F}_\omega = t^2. \tag{6.3}$$

How long should you measure to get the best precision?

(d) Consider now that noise will cause the qubit to decay to its ground state with relaxation time T_1, such that p_1 is modified to

$$p_1 = \sin^2(\frac{\omega t}{2})e^{-t/T_1}. \tag{6.4}$$

Compute the Fisher information in this case. What is the optimal measurement time now?

Problem 6.2: Rabi measurement with non-zero free Hamiltonian

Consider a Rabi measurement on a single qubit consisting of *(i)* initialisation in $|0\rangle$, *(ii)* evolution for time t under the Hamiltonian $\hat{H} = \frac{1}{2}\omega_x\hat{\sigma}_x + \frac{1}{2}\omega_z\hat{\sigma}_z$ and, *(iii)* a projective measurement in the z-basis, $\{|0\rangle, |1\rangle\}$.

(a) Show that the probability for the outcome corresponding to $|1\rangle$ is given by

$$p_1 = \frac{\omega_x^2}{\omega_x^2 + \omega_z^2}\sin^2(\frac{1}{2}\sqrt{\omega_x^2 + \omega_z^2}\, t). \tag{6.5}$$

(b) Plot the corresponding Fisher information $\mathcal{F}_{\omega_x}$ with respect to the transverse frequency ω_x as a function of time, experimenting with different regimes for the frequencies ($\omega_x < \omega_z$, $\omega_x \approx \omega_z$, $\omega_x > \omega_z$).

(c) Argue that the scaling of the optimal $\mathcal{F}_{\omega_x}$ with time is quadratic, t^2.

Problem 6.3: NV thermal populations

The state of a system with Hamiltonian $\hat{H}$ in thermal equilibrium with a bath at temperature T is a mixed state known as a Gibbs state

$$\tau = \frac{1}{Z} e^{-\hat{H}/k_B T}, \tag{6.6}$$

where k_B is Boltzmann's constant and the normalisation factor Z is called the partition function. In particular, for a qubit with energy splitting ϵ, we have

$$\tau = \frac{1}{1 + e^{-\frac{\epsilon}{k_B T}}} \begin{pmatrix} 1 & 0 \\ 0 & e^{-\frac{\epsilon}{k_B T}} \end{pmatrix}. \tag{6.7}$$

The zero-field splitting of an NV center corresponds to angular frequency $\omega = 2.88\,\text{GHz}$.

1. At room temperature at thermal equilibrium, what is the ratio of the probabilities to find the NV qubit in the excited and ground states?
2. What does the temperature need to be for the fraction of population in the ground state to be 99% ?